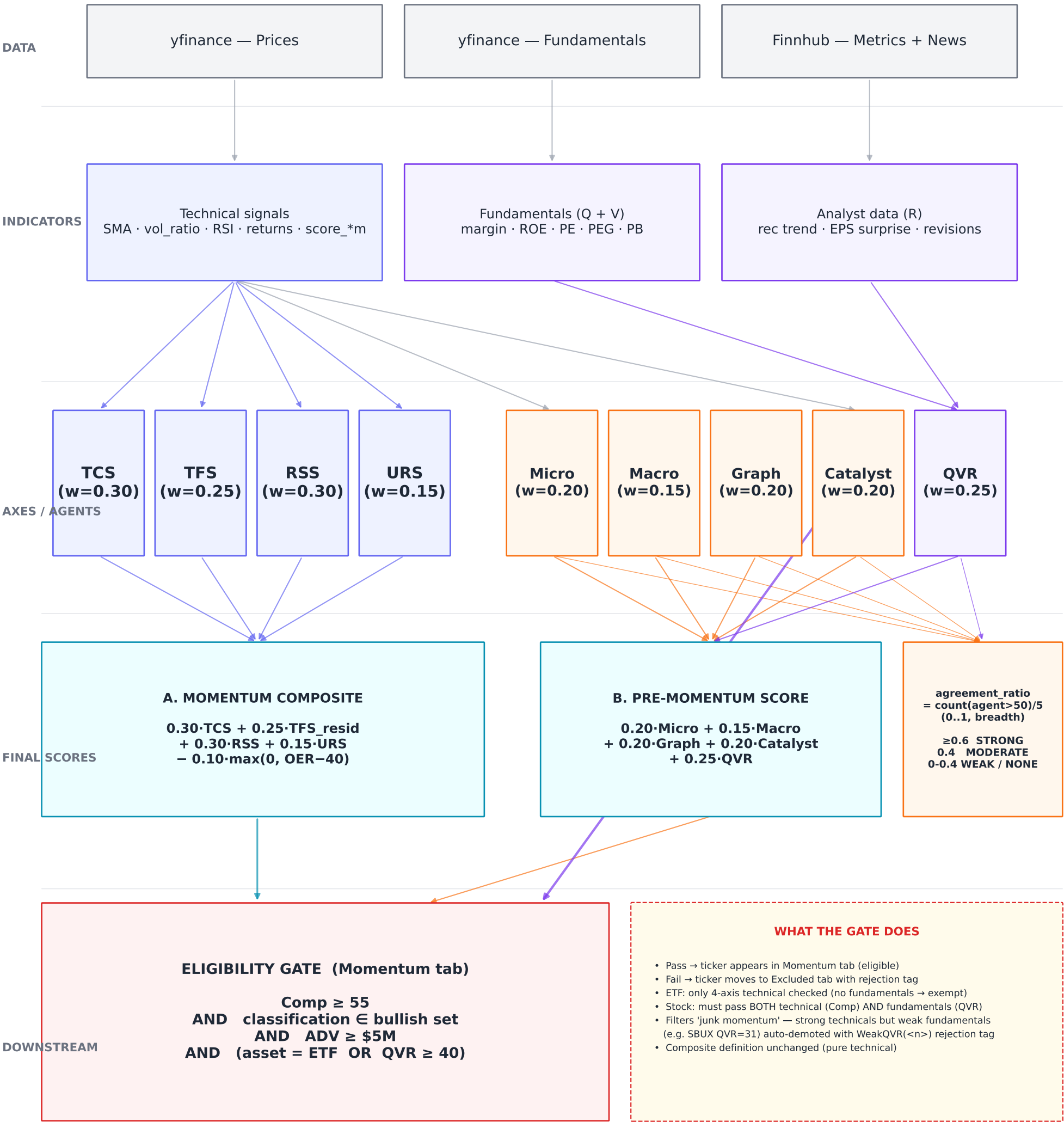


Score System — Dependency Graph

Momentum Composite · Pre-Momentum Score · QVR Eligibility Gate



QVR plays dual role: feeds Pre-Momentum Score (B) AND acts as Eligibility Gate filter · QVR is the only Pre-Mom agent fully orthogonal to Composite (correlation ≈ 0)



Composite path



Pre-Mom path



QVR (dual role)



Final scores



Downstream

Layer 1 + 2 — Data Sources & Indicators

Where every input comes from and how raw data is processed

Layer 1 — Raw Data Sources

All system inputs come from 3 external services, refreshed daily by separate batch scripts.

yfinance — Prices Daily OHLCV for 770 tickers, 5 years history. Computed from this: • SMA20 / SMA50 / SMA200 + slopes • RSI (14-day) • vol_ratio_3d_10d • Returns: 5d / 21d / 63d / 12-1M • Trend age, sma distance	yfinance — Fundamentals Per-ticker accounting + analyst data. Fields used: • trailing PE, forward PE, PEG • gross / operating margins • ROE, debt/equity • EPS estimates (4 quarters) • Recommendations summary • Price targets Limitation: thin coverage for non-US.	Finnhub — Metrics + News US-listed only (free tier). Fields used: • 70+ ratios in single API call • Monthly recommendation history (4 months → bullish_change_3m) • Quarterly EPS surprises (→ beat_rate, avg_surprise) • Company news + timestamps Used to enrich US tickers only.
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Layer 2 — Derived Indicators

Raw data is processed into 3 indicator groups before scoring. Each group flows to specific axes/agents.

Technical signals Fed by: yfinance Prices. Specific computations: • SMA distances: (price-SMA)/SMA • SMA slopes (1-month linear fit) • Trend age (days above SMA50) • vol_ratio_3d_10d • RSI14 (Wilder smoothing) • Returns at 5/21/63/126/252d • 12-1M: prior 11-month return, skipping most recent month Feeds: TCS, TFS, RSS, URS axes; Microstructure / Catalyst.	Fundamentals (Q + V) Fed by: yfinance + Finnhub. QUALITY block: • gross_margin • operating_margin • ROE / ROA • debt/equity, current ratio VALUE block: • forward PE • PEG ratio • price/book (PB) Source priority: • US: Finnhub > yfinance fallback • KR / intl: yfinance only Feeds: QVR Agent (Q + V).	Analyst data (R) Fed by: Finnhub recs + EPS. Specific signals: • bullish_change_3m = bull_ratio(now) - (3m ago) • eps_beat_rate = #beats / #quarters • eps_surprise_avg (% avg) • EPS revision counts (up/down 30d/7d) • Net price-target upside Why this is most leading: Analysts adjust forecasts BEFORE the stock price reacts. Feeds: QVR Agent (R component).
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DAILY UPDATE FLOW

- | | |
|-----------------------------|---|
| 1. price_discovery.py | – runs full scan, produces .scan_cache.pkl (technical signals) |
| 2. fundamentals_pipeline.py | – fetches yfinance fundamentals for all 770 tickers |
| 3. finnhub_fundamentals.py | – enriches cache with Finnhub data for US tickers (~620 of 770) |
| 4. api.py | – loads cache, computes QVR scores, applies Eligibility Gate |

TCS · TFS · RSS · URS — each scored 0-100, combined into Momentum Composite

TCS — Trend Continuation Score

weight 0.30

QUESTION

Is the existing uptrend established and persistent?
(Are we already IN a confirmed trend?)

INPUTS

- SMA20 / SMA50 / SMA200 distance from price
- Slope of each SMA over the last month
- Trend age (consecutive days above SMA50)
- Long-horizon weight 60% > short-horizon 40%

WHAT THE SCORE MEANS

0-30 : trend not established / sideways
30-55 : forming or weak trend
55-75 : confirmed trend
75-100: established, persistent trend

EXAMPLE

NVDA above SMA50 for 60+ days, all SMAs sloping up, price 30% above SMA200 → TCS ~ 80-100.

TFS — Trend Formation Score

weight 0.25

QUESTION

Is a NEW trend forming RIGHT NOW?
(Are we at the start of something?)

INPUTS

- SMA20 / SMA50 breakout strength + freshness
- vol_ratio_3d_10d (3-day vs 10-day avg volume)
- 20-day high proximity
- Slope reversal (was negative, now positive)

WHAT THE SCORE MEANS

0-30 : no breakout / range-bound
30-55 : early signs / weak breakout
55-75 : confirmed fresh breakout
75-100: very recent strong breakout w/ volume

EXAMPLE

Ticker just crossed above SMA50 with 1.8x volume, price near 20-day high → TFS ~ 75-90.

RSS — Relative Strength Score

weight 0.30

QUESTION

How does this ticker rank against the universe?
(Is it outperforming peers?)

INPUTS

- 5d / 21d / 63d / 12-1M return percentile (vs all 770 tickers, cross-sectional rank)
- sma20_slope, vol_ratio percentile
- Long-horizon 65% > short 35%
- 12-1M (Jegadeesh-Titman 1993): prior year minus most recent month

WHAT THE SCORE MEANS

Pure cross-sectional ranking.
0-30 : bottom quartile of universe
55-75 : top half
75-100: top decile

EXAMPLE

Even in a bear market, the best-performing stocks still get RSS 90+ (purely relative).

URS — Underreaction Score

weight 0.15

QUESTION

Is the market slow to digest new information?
(Behavioral overlay on top of pure technicals.)

INPUTS

- LeadLag: peer category avg ret - own ret_63d
- AttnGap: vol_ratio pctile - ret_5d pctile
- Drift: post-event price drift (PEAD proxy)
- Dispersion: cross-sectional std of returns (AQR, Hong-Stein 1999)

WHAT THE SCORE MEANS

0-30 : market has fully reacted / efficient
55-75 : moderate underreaction
75-100: strong attention/price gap (high catch-up potential)

EXAMPLE

Sector peers all up 20%, this ticker only up 5% with rising volume → URS ~ 70-85.

How the 4 axes combine into MOMENTUM COMPOSITE

Composite = 0.30 · TCS + 0.25 · TFS + 0.30 · RSS + 0.15 · URS

TCS + RSS get the highest weight (established + relative strength). URS is a behavioral correction — never dominates.
Composite ≥ 55 is the eligibility threshold (combined with classification + ADV + QVR gate on Stocks).

Each scored 0-100, combined into Pre-Momentum Score (forward-looking signal)

1. Microstructure weight 0.20 DETECTS Pre-breakout structural patterns — low volatility setup, accumulation, RSI compression. SUB-SIGNALS • vol_compression • accumulation_pattern (TFS building while TCS still low) • structural_divergence (structural_q > composite) • volume_regime • range_contraction (RSI ≈ 50) WHY IT'S LEADING Low vol + tight range often precedes breakouts. Inputs use TCS/TFS in INVERSE pattern (low TCS + building TFS = pre-momentum).	2. Macro Regime weight 0.15 DETECTS Top-down regime + rotation + Risk/Style/Region + Bottom-up ETF. SUB-SIGNALS • sector_rotation (0.20) • cross_asset (0.10) • category_breadth (0.20) • relative_improvement (0.15) • rotation_alignment (0.20) • etf_parent_signal (0.15) NEW = bottom-up ETF flag aggregation WHY IT'S LEADING etf_parent_signal (Hybrid Phase D): Stock = weighted avg of parent ETF divergence flags (STEALTH/HEALTHY/WRAPPER/NARROW). ETF = own breadth. Forward-looking bottom-up signal.	3. Graph Relational weight 0.20 DETECTS Theme leadership signals from the GraphRAG knowledge graph. SUB-SIGNALS • peer_lead (% of theme peers above 55) • theme_breadth • leader_lagger_gap (max peer composite - own) • community_momentum WHY IT'S LEADING Uses graph engine's community structure. If theme leaders are running but this ticker is lagging, it's a catch-up candidate.
4. Catalyst weight 0.20 DETECTS Quantitative proxies for catalyst events (no direct news / options data on free tier). SUB-SIGNALS • momentum_acceleration: rss_short - rss_long (short-term RS picking up) • strategy_agreement: long_count / total (8 hedge strategies' consensus) • score_trajectory: score_1w > score_1m > score_3m • reversal_risk_check: reversal_pctile (safety) WHY IT'S LEADING Mostly DELTA/CHANGE measures rather than levels — captures acceleration. Some overlap with Composite (uses score_*m and RSS) is by design: it's the 'building momentum' signal.	5. QVR (Quality + Value + Revision) weight 0.25 DETECTS Fundamentals dimension — only Pre-Mom agent fully orthogonal to Composite (corr ≈ 0). SUB-SIGNALS • Q (30%): margin + ROE percentile (Q_WEIGHTS 0.40 / 0.30 / 0.30) • V (20%): inverse PE/PEG/PB pctile (cheap = high) • R (50%): EPS revisions + bullish_change_3m + eps_beat_rate + eps_surprise_avg + target upside → R weighted highest (Chan-Jegadeesh 1996) WHY IT'S LEADING Independent of price/composite — captures fundamental quality + analyst sentiment momentum BEFORE technicals confirm. Also acts as Eligibility Gate filter (QVR ≥ 40).	

How the 5 agents combine into PRE-MOMENTUM SCORE

Pre-Mom = 0.20·Micro + 0.15·Macro + 0.20·Graph + 0.20·Catalyst + 0.25·QVR

QVR has the highest single weight (0.25) because it's the only fully orthogonal source of forward information.
agreement_ratio = count(agent_score > 50) / 5 (sidecar — measures BREADTH instead of magnitude).

Goal: detect tickers BEFORE Composite confirms momentum — solve the 'after-the-fact' detection problem.

How axes/agents become actionable scores + the 3x3 classification matrix

A. MOMENTUM COMPOSITE (0-100)

base = 0.30·TCS + 0.25·TFS_resid
+ 0.30·RSS_hybrid + 0.15·URS
Composite = base - 0.10·max(0, OER-40)

TFS_resid = TFS residualized vs TCS
(removes TCS-TFS info overlap; same in SVE)
RSS_hybrid = 0.6·within_sector + 0.4·universe
(sector beta correction; sector n>=8)
OER penalty: linear from 40+ (max -6 pts)

Eligibility threshold: >= 55

B. PRE-MOMENTUM SCORE (0-100)

Pre-Mom = 0.20·Micro + 0.15·Macro*
+ 0.20·Graph + 0.20·Catalyst
+ 0.25·QVR

* Macro Regime Agent — 6 sub-signals:
sector_rotation (0.20), cross_asset (0.10),
category_breadth (0.20), rel_improvement (0.15),
rotation_alignment (0.20) [Phase 2C],
etf_parent_signal (0.15) [Hybrid Phase D, NEW]

Purpose: 'Where momentum WILL BE.'
Forward-looking bottom-up signal integrated.

agreement_ratio (sidecar to Pre-Momentum Score, 0..1)

Formula: agreement_ratio = count(agent_score > 50) / 5
Counts HOW MANY of the 5 agents fire simultaneously. Captures BREADTH; PM Score captures MAGNITUDE.

Tier mapping:
≥ 0.6 STRONG — 3+ agents firing → PREPARE / STRONG PREPARE
0.4-0.6 MODERATE — 2 agents firing → WATCH
0-0.4 WEAK — 1 agent firing → TRACK
0 NONE — no agents above 50 → IGNORE

Why both? Case A: QVR=95, others=20 → PM=39 / agree=0.2 (narrow)
Case B: all 5 agents = 55 → PM=55 / agree=1.0 (broad — more reliable)

3 × 3 Classification Matrix (also derived from Composite axes)

Each ticker is also classified into 1 of 9 cells (short × long term direction). Categorical descriptor used for filtering and overrides.

	Long: UP	Long: FLAT	Long: DOWN
Short: UP	CONTINUATION	RECOVERY	COUNTER_RALLY
Short: FLAT	CONSOLIDATION	NEUTRAL	FADING
Short: DOWN	PULLBACK	WEAKENING	DOWNTREND

Overrides (applied AFTER the matrix):

- OVEREXTENDED = OER >= 60 on a bullish cell (warns of mean-reversion risk)
- FORMATION = rapid short-term breakout from a low base (overrides RECOVERY)
- EXHAUSTING = old established trend losing steam (overrides CONTINUATION when age + slope decay)
- CYCLE_PEAK = ret_36_12m percentile >= 85 + 12M momentum declining (long bull cycle peak)
- LAGGING_CATCHUP = sector-lagging name with broad sector strength (catch-up candidate)
- * Phase 3B (api.py): Regime-aware CYCLE_PEAK early promotion
 - Risk-Off + cyclical ticker + OVEREXTENDED + OER>=70 -> CYCLE_PEAK
 - Risk-On + defensive ticker + OVEREXTENDED + OER>=70 -> CYCLE_PEAK (anomaly)
- * Sticky FLAT hysteresis: prev=FLAT -> entry threshold * 1.3 (stricter)
 - Removes bi-weekly NEUTRAL <-> CONSOLIDATION <-> RECOVERY <-> FADING flip noise
 - Bullish set (used by Eligibility Gate): CONTINUATION · FORMATION · RECOVERY · OVEREXTENDED · LAGGING_CATCHUP

How scores + classification become actionable filtering and recommendations

Eligibility Gate — 4 conditions (ALL must pass)

Conditions

1. Composite $\geq$ 55

(technical strength threshold)

2. Classification $\in$ bullish set

(CONTINUATION/FORMATION/RECOVERY/OVEREXTENDED/LAGGING_CATCHUP)

3. ADV $\geq$ \$5M

(liquidity floor — avg daily volume in USD)

4. Asset-class branch:

ETF: skip (no fundamentals available — exempt)

Stock: QVR $\geq$ 40 (fundamentals sanity check)

Pass $\rightarrow$ Momentum tab (eligible)

Fail $\rightarrow$ Excluded tab (with rejection tag explaining why)

Rejection Tag Catalog (shown in Excluded tab)

Tag types (multiple can combine, e.g. 'CONTINUATION/WeakQVR(31)')

LowScore

— Composite < 55 (technical strength insufficient)

Liq(X.XM)—ADVbelow5M floor; actual value shown in parens

Downtrend

— classification = DOWNTREND (long+short both DOWN)

CyclePeak

— long bull cycle showing exhaustion signs

Fading

— classification = FADING (short FLAT, long DOWN — losing momentum)

Weakening

— classification = WEAKENING (short DOWN, long FLAT — bearish forming)

Exhausting

— long-established trend losing slope (override of CONTINUATION)

WeakQVR(N)

— Stock with QVR below 40; technical may be strong but fundamentals are weak

(the 'junk momentum' filter — example: SBUX classified CONTINUATION but QVR=31)

Decision Logic (Action recommendation per ticker)

Momentum tab — decideAction()

Inputs: composite, classification, OER, age, signal

Output tags:

BUY

composite $\geq$ 65 + STRONG_LONG signal

HOLD

composite $\geq$ 55, signal LONG/NEUTRAL

TRIM

OER $\geq$ 60 (overextended)

HEDGE

OER $\geq$ 70 + classification OVEREXTENDED

EXIT

classification DOWNTREND or CYCLE_PEAK

WATCH

composite 55-65, signal NEUTRAL

Sorting: rank 1 (most actionable BUY)
rank 10 (EXIT / HEDGE)

Pre-Momentum tab — decidePMAction()

Inputs: pre_momentum_score, agreement_ratio, age, classification

Output tags:

STRONG PREPARE

agreement $\geq$ 0.6 + PM $\geq$ 75 + age $\geq$ 14d

PREPARE

agreement $\geq$ 0.6 + PM $\geq$ 65

WATCH (Active)

agreement $\geq$ 0.4 + PM $\geq$ 60 + 3+ catalysts

WATCH

agreement $\geq$ 0.4 + PM $\geq$ 55

TRACK

weak agreement but PM $\geq$ 50

STAGNANT

age $\geq$ 80d but agreement not strong

IGNORE

weak agreement + low score

Risk override: FADING/WEAKENING classification
 $\rightarrow$ IGNORE (or 'WATCH (caution)' if PM still strong)

Tab Routing + Phase 1-3 Macro Context Tags (api.py post-load)

Momentum tab : passes ALL 4 gate conditions -> current actionable holdings
Excluded tab : bearish classification OR demoted by WeakQVR -> monitor for exit / avoidance
Pre-Momentum tab : not yet eligible BUT pre_momentum_score signals forming -> watchlist / preparation

-- Macro Context Tags (api.py post-load) --
* Phase 1 : cyclical_tag . style_tilt . region . industry_group (+ Biotech/Telecom industry refinement)
* Phase 1G : ve_obs tag injection -> /api/validation segmented hit rates (cyc/style/region)
* Phase 2C : MacroRegimeAgent rotation_alignment sub-signal (weight 0.20)
* Phase 2D : per-ticker rotation_long / rotation_short scores
* Phase 3B : Regime-aware CYCLE_PEAK early promotion (Risk-Off + cyclical + OVEREXTENDED + OER>=70)
End of document. · Source: pre_momentum.py · price_discovery.py · qvr_agent.py · api.py · etf_holdings_pipeline.py

-- Hybrid Bottom-up (ETF constituent integration) --
* Phase A : etf_holdings_pipeline.py fetches top-10 holdings via yfinance (70 ETFs cached)
* Phase A+B : ETF sidecar metrics (constituent_breadth_mom, weighted_comp, coverage, leader_gap)
+ divergence_flag = {HEALTHY_TREND, NARROW_RALLY, STEALTH_STRENGTH, WRAPPER_DRAG, NEUTRAL}
* Phase C : MarketCommentaryTab ETF Hybrid Health Card (per-ETF 8-col table + flag badges)
* Phase D : parent_etf_signal -> MacroRegimeAgent etf_parent_signal sub-signal (weight 0.15)
Stock = weighted avg of parent ETF flags; ETF = own constituent breadth
* Phase E : Conviction Picks BuyScore += ETF flag bonus (STEALTH +10, HEALTHY +6, NARROW -4)